COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 2 : Session – 3 : SLO – 2 | Multiple Bus Structures & Expansion Bus Types
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Section A – Match the Following (Answers)

- 1. ISA → D. Industry Standard Architecture
- 2. Multiple Bus Architecture → A. Uses interleaved memory banks to improve performance
- 3. PCI → C. Peripheral Component Interconnect
- 4. AGP → B. Handles video-specific data transfers
- 5. PCMCIA → E. PC memory card interface

Section B – Fill in the Blanks

- A MULTIPLE BUS is used to overcome the bandwidth limitations of a single bus system.
- In a multiple bus system, memory is divided into multiple BANKS (modules) to allow parallel access.
- To handle simultaneous memory requests, a BUS ARBITRATION policy is needed.
- The AGP bus is specifically designed to accelerate graphics rendering.
- EISA serves as an extended version of the ISA expansion bus.

Section C – Activity Task

- Task 1 (described): Processors P1…PN, each with a private cache, connect to multiple buses B1…BB; the buses connect to interleaved memory banks M1…MM; hardware arbiters at each bus resolve contention.
- Task 2a – Interleaved memory: consecutive addresses are spread across different banks, so while one bank is being accessed another can start a new access; overlapping accesses raise effective memory bandwidth.
- Task 2b – More processors than buses: contention grows; some processors must wait for a free bus, so throughput saturates at the number of buses and arbitration delays increase.
- Task 2c – Hardware arbiters: several processors may request the same bus in the same cycle; arbiters grant access according to priority/fairness rules so that transfers do not collide and every requester is eventually served.

Section D – True or False

- In a multiple bus system, a processor is connected to only one bus. — FALSE
- Interleaved memory allows simultaneous access to different memory banks. — TRUE
- Bus allocation in a multiple bus system is handled by software scheduling. — FALSE (hardware arbiters)
- The PCI bus is an expansion bus used for connecting external hardware. — TRUE
- Even if some paths fail in a multiple bus architecture, the system can still operate. — TRUE